0.1 Diagrams for Problems 3–5

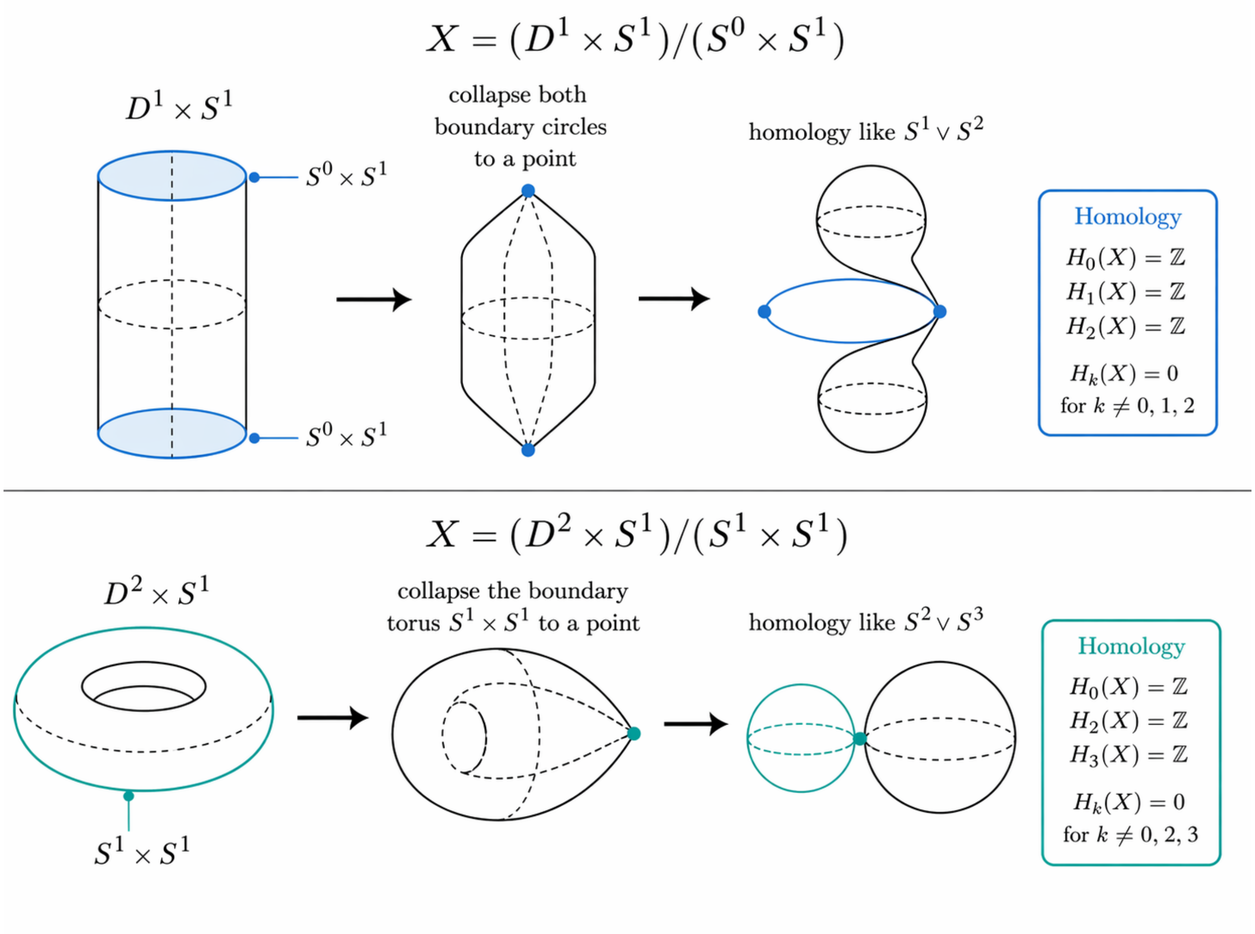

Figure 1: Low-dimensional quotient examples for Problem 3. The quotient $(D^1 \times S^1)/(S^0 \times S^1)$ behaves homologically like $S^1 \vee S^2$, while $(D^2 \times S^1)/(S^1 \times S^1)$ behaves homologically like $S^2 \vee S^3$.

Cellular chain complex of L^3_p (with $\mathbb{Z}$-coefficients)

$$0 \longrightarrow \underset{\substack{\text{3-cells}\\ \text{(dim 3)}}}{\mathbb{Z}} \xrightarrow{0} \underset{\substack{\text{2-cells}\\ \text{(dim 2)}}}{\mathbb{Z}} \xrightarrow{\times p} \underset{\substack{\text{1-cells}\\ \text{(dim 1)}}}{\mathbb{Z}} \xrightarrow{0} \underset{\substack{\text{0-cells}\\ \text{(dim 0)}}}{\mathbb{Z}} \longrightarrow 0$$

$$\boxed{\pi_1(L^3_p) = \mathbb{Z}/p}$$

Homology groups of L^3_p (with $\mathbb{Z}$-coefficients)	
$H_0(L^3_p)$	$\mathbb{Z}$
$H_1(L^3_p)$	$\mathbb{Z}/p$
$H_2(L^3_p)$	0
$H_3(L^3_p)$	$\mathbb{Z}$

Figure 2: The lens space L^3_p as a quotient of S^3, together with its cellular chain complex and homology groups.

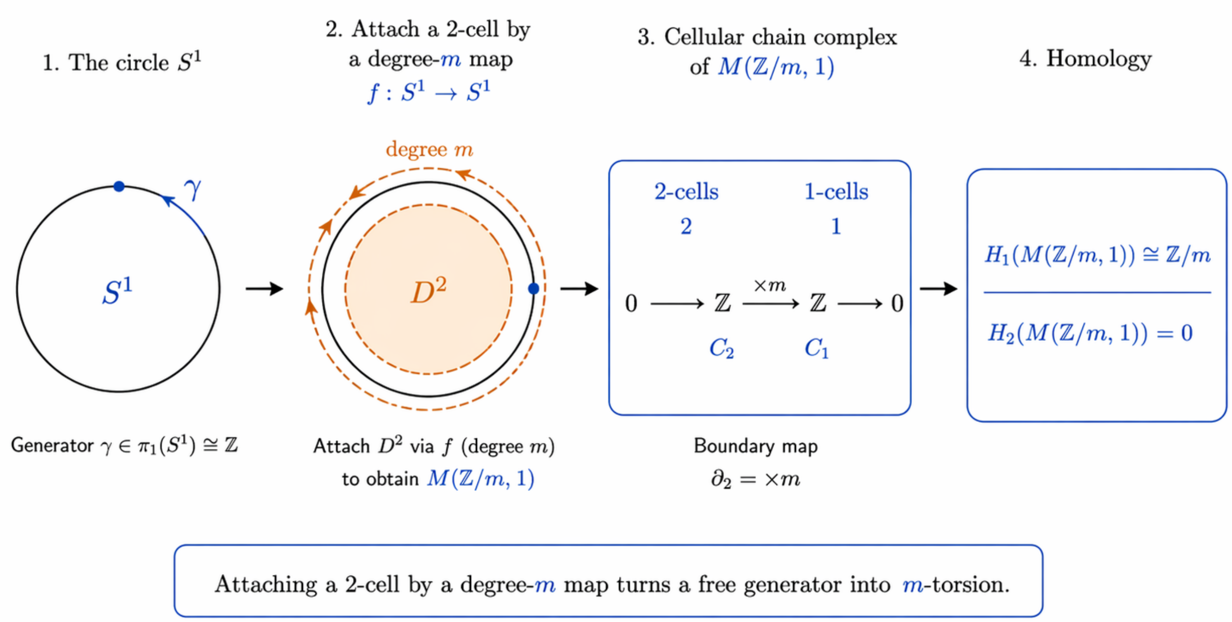

Figure 3: Construction of the Moore space $M(\mathbb{Z}/m, 1)$ by attaching a 2-cell by a degree-m map, and the resulting cellular homology.

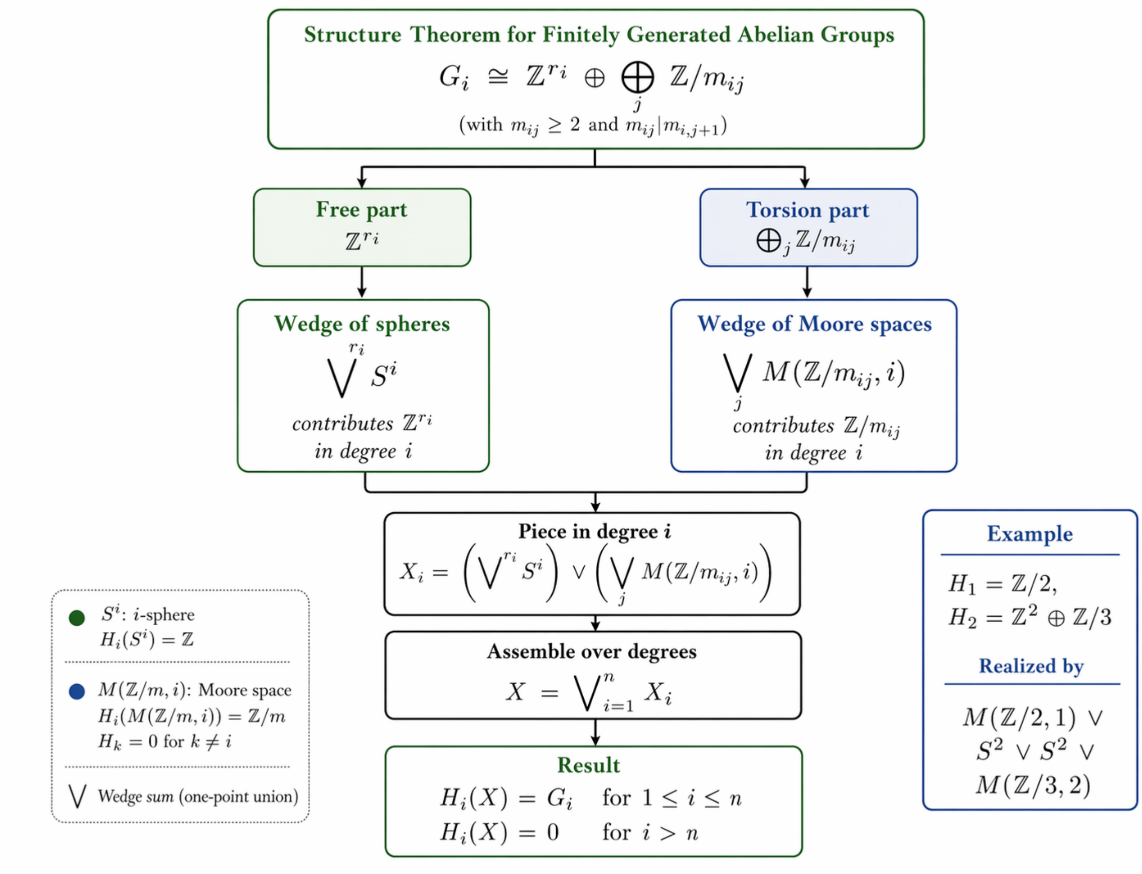

Figure 4: Realization of prescribed finitely generated abelian homology groups using wedges of spheres and Moore spaces, as in Problem 5.